

Generating and Reranking Diverse Answers in Video Question Answering: TREC2025 Video Question Answering Shared Task Description

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Abstract

*Video-based question answering (Video QA) remains a core challenge in multimodal understanding due to the complex temporal and causal dynamics of video data. We propose a listwise learning framework that improves semantic precision and ranking consistency in answer generation. Given a video-question pair, a base multimodal model first generates multiple candidate answers, which are then reranked using a model trained with a novel Masked Pointer Cross-Entropy Loss with Rank Weights. This objective integrates pointer-based candidate selection, rank-dependent weighting, and masked cross-entropy under vocabulary restriction, enabling stable and interpretable listwise optimization. By bridging generative modeling with discriminative ranking, our method produces coherent, fine-grained answer lists. Experiments on the TREC2025 VQA dataset show consistent improvements in accuracy and ranking stability, especially for questions requiring temporal reasoning and semantic disambiguation.*¹

1. Introduction

The rapid improvement of vision-language models (VLMs) has advanced multimodal understanding across a variety of tasks in text, image, audio, and video [10, 39, 40]. Despite this, video understanding remains particularly challenging; unlike static visual scenes, videos require models to integrate spatial, auditory, and temporal cues to construct coherent representations over time [18, 24, 25, 35, 53]. Developing systems capable of such reasoning is a key step toward general multimodal intelligence.

Within this context, Video Question Answering (Video QA) is a fundamental benchmark for assessing multimodal reasoning. Video QA involves generating natural language

answers to questions grounded in video content, requiring models to align visual and textual semantics while maintaining temporal and contextual coherence [15, 29, 50]. In the TREC2025 VQA task, systems must generate multiple plausible answers per query and return them as a ranked list. This setting continues to be difficult for current VLMs, which often struggle to generate diverse, well-ranked answer candidates and to maintain stable orderings in open-ended settings [8, 11, 28].

To address these limitations, we explore a listwise learning framework for answer generation and ranking. Given a video and an associated question, our system first generates multiple candidate answers using a base VLM, and then reorders them using a reranker. For reranking, we introduce a novel **Masked Pointer Cross-Entropy Loss with Rank Weights**, which integrates three key mechanisms. First, rank-labeled candidate representation explicitly encodes ordinal relations among answer candidates. Next, masked pointer cross-entropy facilitates permutation-consistent optimization through selective supervision of ranked outputs. Finally, rank-dependent weighting enhances the learning signal for higher-priority predictions, while maintaining balanced gradient propagation. Together, these mechanisms allow the model to capture both ranking consistency and linguistic fluency, effectively bridging the gap between generative modeling and discriminative ranking.

Our contributions can be summarized as follows:

1. We propose a generate and rerank setting for producing diverse ranked lists in open-ended Video QA.
2. We introduce **Masked Pointer Cross-Entropy Loss with Rank Weights**, explicitly incorporating ranking information into the optimization process.
3. Extensive experiments demonstrate that our proposed pipeline yields consistent improvements over baseline models across multiple evaluation metrics.

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¹Code: <https://github.com/Charles201428/2025-VQA-AG/tree/research-framework-vqa>

2. Related Work

TREC VQA The Video Question Answering (VQA) Answer Generation (AG) task, organized under the TRECVID benchmark by NIST,² is designed to evaluate the ability of multimodal models to understand and reason about video content. The objective is to automatically generate up to ten ranked textual answers for each video–question pair and participants are provided with a set of short video clips from V3C [approximately 30 seconds each; 31] and corresponding natural language questions. Evaluation of the task is performed using automatic semantic similarity metrics: Semantic Textual Similarity [STS; 5], METEOR [3], and BERTScore [49], along with normalized discounted cumulative gain [nDCG; 14] to assess ranking quality.

Video Understanding Vision Language Models (VLMs) [13, 41, 45, 51] have fundamentally reshaped the paradigm of video understanding and question answering. By integrating powerful language reasoning with visual perception, VLMs can interpret, describe, and reason about video content in a unified manner. Different architectural paradigms—such as video analyzer-based [7, 23, 48], video embedding-based [6, 9], and hybrid fusion frameworks [19, 43]—connect visual encoders with LLM backbones through textual or feature-level alignment, enabling end-to-end multimodal comprehension.

In Video Question Answering (VQA), VLMs demonstrate strong capabilities in temporal reasoning, causal inference, and commonsense understanding [16, 24, 42]. Through in-context learning [4] and chain-of-thought prompting [36], these models can answer complex questions and perform fine-grained event reasoning [12, 25, 30].

Reranking LLM-based reranking has recently become a core component in modern information retrieval [1, 52]. Following an initial retrieval stage that gathers candidate passages or answers, large language models are increasingly used to perform reranking, leveraging their semantic understanding and reasoning abilities to refine relevance estimation. Early studies adopt point-wise reranking [22, 27, 37], where the model independently scores each query–document or query–answer pair. More recent approaches explore list-wise reranking [21, 44, 46], where the model jointly considers a set of candidates to capture comparative relevance and global ranking consistency.

Beyond purely textual settings, LLM rerankers have also been extended to multimodal information retrieval, aligning text with visual or audio evidence for support judgments across modalities [20, 26, 32, 33, 47]. In question answering tasks, these models can rerank answer candidates by as-

sessing textual entailment and evidence support between the generated responses and passages, images, or videos.

3. Method

Our approach consists of two main components: (1) a **generator** that produces multiple candidate answers given a query, and (2) a **reranker** that refines these candidates into a ranked list using a masked pointer-based objective. An overview of the full pipeline is shown in Figure 1.

3.1. Generator

The generator module is responsible for producing a diverse set of candidate answers for each input query q . For each query, we generate N candidates $\{a_1, a_2, \dots, a_N\}$, which are subsequently used as input to the reranker.

To encourage diversity, we employ a collection of prompt templates that differ in phrasing, structure, and reasoning cues. Instead of tuning model parameters or sampling hyperparameters, we rely on prompt variation as a lightweight and reproducible way to obtain diverse candidates. These prompts are empirically evaluated on a validation set, and the template that yields the best reranking performance is selected and fixed for both training and evaluation. This setup ensures that the generation process remains deterministic and reproducible, while the resulting candidate pool maintains sufficient diversity.

3.2. Reranker

The reranker aims to reorder the generated candidates $\{a_1, a_2, \dots, a_N\}$ into a coherent ranked list that aligns with ground-truth relevance. It is trained using a **Masked Pointer Cross-Entropy Loss with Rank Weights**, which integrates three key mechanisms: (1) Rank-Labeled Candidate Representation, (2) Masked Pointer Cross-Entropy, and (3) Rank-Dependent Weighting.

3.2.1. Rank-Labeled Candidate Representation

For each query q with its reference answer a^* , we first compute the BERTScore between each generated candidate a_i and a^* . Candidates are then ordered to form a ranked sequence in a fixed way:

```
<R1> = <CAND_...>
<R2> = <CAND_...>
...
<RN> = <CAND_...>
<ENDLIST>
```

Each token in this sequence represents either a rank marker ($\langle R_n \rangle$), a candidate pointer ($\langle \text{CAND}_m \rangle$), or regular textual content. This rank-labeled sequence serves as the target supervision for the reranker.

²<https://www-nlpir.nist.gov/projects/tv2025/vqa.html>

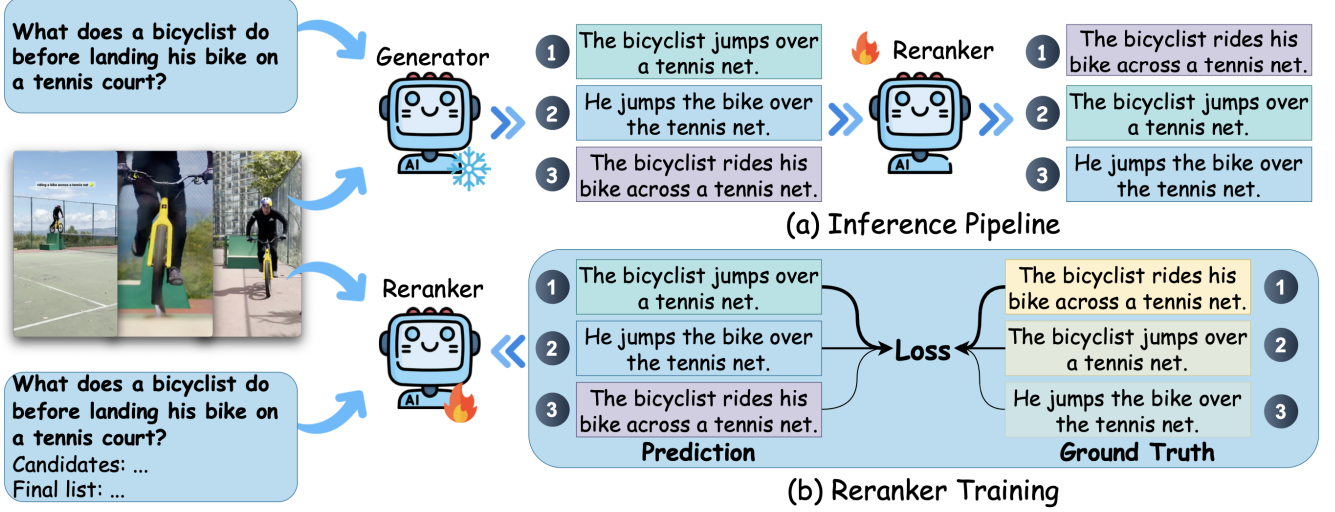


Figure 1. Overview of the proposed generate-rank framework. The framework comprises two key components: a Generator, which produces a set of candidate responses given an input prompt, and a Reranker, which assigns relevance scores to these candidates and orders them accordingly. (a) **Inference Pipeline**: The generator outputs multiple candidate responses, which are subsequently reordered by the reranker based on their predicted relevance to the query. (b) **Reranker Training**: The reranker is trained using ground-truth rankings, where token-level weighting is incorporated into the loss function to emphasize more informative tokens. Darker colors denote higher token weights, indicating a greater contribution to the overall loss.

3.2.2. Masked Pointer Cross-Entropy

The reranker is trained to autoregressively generate the rank-labeled sequence. At each decoding step t , the model produces logits $\mathbf{z}_t \in \mathbb{R}^V$ over the vocabulary. When the target token corresponds to a candidate pointer (e.g., `<CAND_3>`) or the termination token `<ENDLIST>`, the prediction space is restricted to the subset of valid, unselected candidates $\mathcal{C}_k$:

$$P(y^{(k)}|\mathcal{C}_k) = \frac{\exp(z_{y^{(k)}}^{(k)})}{\sum_{j \in \mathcal{C}_k} \exp(z_j^{(k)})}. \quad (1)$$

The corresponding pointer loss is:

$$\mathcal{L}_{\text{ptr}}^{(k)} = -\log P(y^{(k)}|\mathcal{C}_k). \quad (2)$$

For non-pointer tokens (e.g., rank markers or normal text), the model computes standard cross-entropy over the full vocabulary:

$$\mathcal{L}_{\text{text}}^{(t)} = -\log \frac{\exp(z_{y_t}^{(t)})}{\sum_{v=1}^V \exp(z_v^{(t)})}. \quad (3)$$

3.2.3. Rank-Dependent Weighting

To emphasize higher-ranked predictions, we assign a weight w_k to each pointer step corresponding to rank k :

$$w_k = \begin{cases} \alpha_k, & \text{if } k \leq 3, \\ 1, & \text{otherwise,} \end{cases} \quad (4)$$

where $\alpha_k > 1$ are hyperparameters controlling the relative importance of top-ranked predictions. These weights are applied directly to the loss terms at respective rank steps.

3.2.4. Final Objective

The final training objective aggregates all weighted losses:

$$\mathcal{L}_{\text{total}} = \frac{1}{\sum_t w_t} \sum_t w_t \mathcal{L}_t, \quad (5)$$

where $\mathcal{L}_t$ is either $\mathcal{L}_{\text{ptr}}^{(k)}$ or $\mathcal{L}_{\text{text}}^{(t)}$.

This formulation ensures that:

1. candidate tokens are compared only within the subset of valid remaining candidates;
2. higher-ranked predictions receive stronger supervision;
3. gradient flow remains stable without in-place masking.

The proposed **Masked Pointer Cross-Entropy with Rank Weights** thus enables the model to learn fine-grained correspondences between candidate order and predicted likelihood, achieving stable and consistent listwise reranking performance.

4. Experiment

Data and Models The dataset for both training and evaluation is the VQA dataset provided by TREC, with the dev data included as part of the training set. For the **generator**, we evaluate both Qwen2.5-VL-7B [2] and Qwen2.5-Omni-7B [38] to compare their generative capabilities. For the **reranker**, we employ Qwen2.5-VL-7B as the backbone model and fine-tune it specifically for ranking refinement.

Split	Settings	nDCG		BERTScore		METEOR	
		BERTScore Rank	METEOR Rank	Mean	Best	Mean	Best
Dev Data	QwenOmni Generator	0.997	0.772	0.869	0.875	0.145	0.182
	QwenVL Generator	0.995	0.664	0.904	0.905	0.175	0.183
	QwenVL Generator+Reranker	0.998	0.742	0.904	0.916	0.175	0.250
Test Data	QwenVL Generator	0.995	0.708	0.896	0.896	0.157	0.159
	QwenVL Generator+Reranker	0.995	0.718	0.896	0.897	0.156	0.161

Table 1. Comprehensive evaluation of different generator configurations on the video question answering task. Results are reported on both the development and test splits under three metrics: NDCG, BERTScore, and METEOR. QwenVL achieves superior overall performance compared to QwenOmni, particularly in semantic similarity metrics (BERTScore and METEOR). The inclusion of a reranker further refines ranking quality, yielding the most notable gains under the METEOR metric.

4.1. Evaluation Metrics

To assess the quality of ranked answers, we adopt the same metrics as the TREC shared task. Specifically, we report normalized discounted cumulative gain (nDCG) for two metrics, BERTScore [49] and METEOR [3]. We also report both the mean, the average score across all generated answers, and the best, the score corresponding to the top-ranked (first) answer, for BERTScore and METEOR. For our ablations on prompting, we also report ROUGE-L [17] and Semantic Text Similarity [STS; 5].

4.2. Results

The overall results are presented in Table 1. We evaluate two main configurations:

1. **Generator**: a single-stage model that directly generates candidate answers.
2. **Generator + Reranker**: a two-stage framework where the generator produces candidate answers and the reranker refines their ranking.

For the development split, we additionally test the Qwen Omni Generator to investigate the performance of the Omni variant. Although Qwen Omni achieves slightly higher NDCG scores, it performs worse in BERTScore and METEOR. Since our goal is to achieve a balanced improvement across all evaluation metrics, and as shown in [subsection 4.3](#), fine-tuning the generator alone does not substantially enhance semantic alignment, *we ultimately adopt Qwen-VL as the generator in our final system*.

4.3. Prompt Study

To further examine the influence of instruction formulation and fine-tuning strategies, we conduct an ablation study on prompt templates and LoRA-based fine-tuning.³ Each prompt variant represents a distinct instruction template used during generation.

³See [Appendix A](#) for the parameters used in training. The prompt variants can be found in [Appendix B](#).

Prompt	ROUGE-L	METEOR	STS
PROMPT1	0.224	0.222	0.469
PROMPT2	0.334	0.216	0.607
PROMPT3	0.389	0.228	0.631
PROMPT4	0.383	0.227	0.623
PROMPT5	0.407	0.251	0.638
PROMPT6	0.390	<u>0.246</u>	0.620
PROMPT7	0.352	0.245	0.595
PROMPT5 w/ lora	<u>0.394</u>	0.239	<u>0.636</u>

Table 2. Ablation results of different prompt templates and their LoRA variants. Scores are reported under ROUGE-L, METEOR, and STS metrics.

Table 2 reports the raw evaluation results under three metrics: ROUGE-L, METEOR, and STS. Among all prompt variants, Prompt5 consistently achieves the best overall performance across metrics. *Accordingly, we adopt Prompt5 as the final instruction template for the generator in our final system*.

5. Conclusion/Discussion

We conducted extensive experiments on the TREC-VQA dataset to evaluate the effectiveness of our proposed framework. Our study systematically examined the influence of prompt design on the generator’s performance and introduced a weighted reranking mechanism to optimize the ordering of candidate answers. Experimental results demonstrate that well-crafted prompts play a crucial role in producing semantically accurate and contextually relevant responses. Moreover, the proposed **Masked Pointer Cross-Entropy Loss with Rank Weights** substantially enhances ranking consistency and improves the quality of top-ranked answers. We believe that this training paradigm provides valuable insights for future research on rank-list optimization and could serve as a general framework for learning to rank in multimodal generation tasks.

Limitation and Future Work

Given that the current framework is trained exclusively on the TREC dataset, its overall performance may be constrained by the limited diversity and scale of the training data. Enhancing the reranker through exposure to additional datasets could therefore lead to more robust and generalizable results. Similarly, the suboptimal performance of the generator may also be attributed to the restricted amount of training data. Jointly training the generator and reranker on a larger and more diverse corpus could further improve the quality and consistency of generated outputs. Furthermore, the proposed framework can be extended to other promising research directions, such as reinforcement learning-based reasoning mechanisms for QA tasks. For instance, within the GRPO framework [34], the reranker can be incorporated into the rollout phase to assess and rank the generated candidate answers, thereby providing differentiated auxiliary rewards according to their relative quality, rather than depending solely on a verification-based reward signal. This integration could enable a more refined and effective optimization process.

References

- [1] Abdelrahman Abdallah, Bhawna Piryani, Jamshid Mozafari, Mohammed Ali, and Adam Jatowt. How good are llm-based rerankers? an empirical analysis of state-of-the-art reranking models. *arXiv preprint arXiv:2508.16757*, 2025. 2
- [2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhao-hai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. 3
- [3] Satandeep Banerjee and Alon Lavie. Meteor: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005. 2, 4
- [4] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020. 2
- [5] Daniel Cer, Mona Diab, Eneko Agirre, Iñigo Lopez-Gazpio, and Lucia Specia. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *Proceedings of the 11th International Workshop on Semantic Evaluation (SemEval-2017)*, pages 1–14, Vancouver, Canada, 2017. Association for Computational Linguistics. 2, 4
- [6] Lin Chen, Xilin Wei, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Zhenyu Tang, Li Yuan, et al. Sharegpt4video: Improving video understanding and generation with better captions. *Advances in Neural Information Processing Systems*, 37:19472–19495, 2024. 2
- [7] Sihan Chen, Handong Li, Qunbo Wang, Zijia Zhao, Mingzhen Sun, Xinxin Zhu, and Jing Liu. Vast: A vision-audio-subtitle-text omni-modality foundation model and dataset. *Advances in Neural Information Processing Systems*, 36:72842–72866, 2023. 2
- [8] Yangyi Chen, Karan Sikka, Michael Cogswell, Heng Ji, and Ajay Divakaran. Measuring and improving chain-of-thought reasoning in vision-language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 192–210, 2024. 1
- [9] Zesen Cheng, Sicong Leng, Hang Zhang, Yifei Xin, Xin Li, Guanzheng Chen, Yongxin Zhu, Wenqi Zhang, Ziyang Luo, Deli Zhao, et al. Videollama 2: Advancing spatial-temporal modeling and audio understanding in video-llms. *arXiv preprint arXiv:2406.07476*, 2024. 2
- [10] Gheorghe Comanici, Eric Bieber, Mike Schaekermann, Ice Pasupat, Naveen Sachdeva, Inderjit Dhillon, Marcel Blstein, Ori Ram, Dan Zhang, Evan Rosen, et al. Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities. *arXiv preprint arXiv:2507.06261*, 2025. 1
- [11] Chaoyou Fu, Yuhang Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 24108–24118, 2025. 1
- [12] Songhao Han, Wei Huang, Hairong Shi, Le Zhuo, Xiu Su, Shifeng Zhang, Xu Zhou, Xiaojuan Qi, Yue Liao, and Si Liu. Videoespresso: A large-scale chain-of-thought dataset for fine-grained video reasoning via core frame selection. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 26181–26191, 2025. 2
- [13] Hang Hua, Jing Shi, Kushal Kafle, Simon Jenni, Daoan Zhang, John Collomosse, Scott Cohen, and Jiebo Luo. Fine-match: Aspect-based fine-grained image and text mismatch detection and correction. In *European Conference on Computer Vision*, pages 474–491. Springer, 2024. 2
- [14] Kalervo Järvelin and Jaana Kekäläinen. Cumulated gain-based evaluation of ir techniques. *ACM Transactions on Information Systems (TOIS)*, 20(4):422–446, 2002. 2
- [15] Nayoung Kim, Seong Jong Ha, and Je-Won Kang. Video question answering using language-guided deep compressed-domain video feature. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 1708–1717, 2021. 1
- [16] Jie Lei, Licheng Yu, Mohit Bansal, and Tamara L Berg. Tvqa: Localized, compositional video question answering. *arXiv preprint arXiv:1809.01696*, 2018. 2
- [17] Chin-Yew Lin. ROUGE: A package for automatic evaluation of summaries. In *Text Summarization Branches Out*, pages 74–81, Barcelona, Spain, 2004. Association for Computational Linguistics. 4
- [18] Jingyang Lin, Jialian Wu, Ximeng Sun, Ze Wang, Jiang Liu, Yusheng Su, Xiaodong Yu, Hao Chen, Jiebo Luo, Zicheng

- Liu, and Emad Barsoum. Unleashing hour-scale video training for long video-language understanding, 2025. 1
- [19] Kevin Lin, Faisal Ahmed, Linjie Li, Chung-Ching Lin, Ehsan Azarnasab, Zhengyuan Yang, Jianfeng Wang, Lin Liang, Zicheng Liu, Yumao Lu, et al. Mm-vid: Advancing video understanding with gpt-4v (ision). *arXiv preprint arXiv:2310.19773*, 2023. 2
- [20] Jingzhou Liu, Wenhu Chen, Yu Cheng, Zhe Gan, Licheng Yu, Yiming Yang, and Jingjing Liu. Violin: A large-scale dataset for video-and-language inference, 2020. 2
- [21] Wenhan Liu, Xinyu Ma, Weiwei Sun, Yutao Zhu, Yuchen Li, Dawei Yin, and Zhicheng Dou. Reasonrank: Empowering passage ranking with strong reasoning ability, 2025. 2
- [22] Xueguang Ma, Liang Wang, Nan Yang, Furu Wei, and Jimmy Lin. Fine-tuning llama for multi-stage text retrieval, 2023. 2
- [23] Ziyu Ma, Chenhui Gou, Hengcan Shi, Bin Sun, Shutao Li, Hamid Rezaatofighi, and Jianfei Cai. Drvideo: Document retrieval based long video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 18936–18946, 2025. 2
- [24] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Shahbaz Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv preprint arXiv:2306.05424*, 2023. 1, 2
- [25] Alexander Martin, Reno Kriz, William Gantt Walden, Kate Sanders, Hannah Recknor, Eugene Yang, Francis Ferraro, and Benjamin Van Durme. Wikivideo: Article generation from multiple videos, 2025. 1, 2
- [26] Alexander Martin, William Walden, Reno Kriz, Dengjia Zhang, Kate Sanders, Eugene Yang, Chihsheng Jin, and Benjamin Van Durme. Seeing through the mirage: Evaluating multimodal retrieval augmented generation, 2025. 2
- [27] Rodrigo Nogueira, Wei Yang, Kyunghyun Cho, and Jimmy Lin. Multi-stage document ranking with bert, 2019. 2
- [28] Jean Park, Kuk Jin Jang, Basam Alasaly, Sriharsha Mopidevi, Andrew Zolensky, Eric Eaton, Insup Lee, and Kevin Johnson. Assessing modality bias in video question answering benchmarks with multimodal large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 19821–19829, 2025. 1
- [29] AJ Piergiovanni, Kairo Morton, Weicheng Kuo, Michael S Ryoo, and Anelia Angelova. Video question answering with iterative video-text co-tokenization. In *European Conference on Computer Vision*, pages 76–94. Springer, 2022. 1
- [30] Long Qian, Juncheng Li, Yu Wu, Yaobo Ye, Hao Fei, Tat-Seng Chua, Yueting Zhuang, and Siliang Tang. Momen-tor: Advancing video large language model with fine-grained temporal reasoning. *arXiv preprint arXiv:2402.11435*, 2024. 2
- [31] Luca Rossetto, Heiko Schuldt, George Awad, and Asad A Butt. V3c—a research video collection. In *International Conference on Multimedia Modeling*, pages 349–360. Springer, 2018. 2
- [32] Kate Sanders and Benjamin Van Durme. Bonsai: Interpretable tree-adaptive grounded reasoning, 2025. 2
- [33] Kate Sanders, Reno Kriz, Anqi Liu, and Benjamin Van Durme. Ambiguous images with human judgments for robust visual event classification, 2022. 2
- [34] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, YK Li, Yang Wu, et al. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. 5
- [35] Yan Shu, Zheng Liu, Peitian Zhang, Minghao Qin, Junjie Zhou, Zhengyang Liang, Tiejun Huang, and Bo Zhao. Video-xl: Extra-long vision language model for hour-scale video understanding. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 26160–26169, 2025. 1
- [36] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35:24824–24837, 2022. 2
- [37] Orion Weller, Kathryn Ricci, Eugene Yang, Andrew Yates, Dawn Lawrie, and Benjamin Van Durme. Rank1: Test-time compute for reranking in information retrieval, 2025. 2
- [38] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. Qwen2.5-omni technical report, 2025. 3
- [39] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, et al. Qwen2.5-omni technical report. *arXiv preprint arXiv:2503.20215*, 2025. 1
- [40] Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, Jinzheng He, Yuxuan Wang, Xian Shi, Ting He, Xinfa Zhu, et al. Qwen3-omni technical report. *arXiv preprint arXiv:2509.17765*, 2025. 1
- [41] Shiyu Xuan, Qingpei Guo, Ming Yang, and Shiliang Zhang. Pink: Unveiling the power of referential comprehension for multi-modal llms. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 13838–13848, 2024. 2
- [42] Antoine Yang, Antoine Miech, Josef Sivic, Ivan Laptev, and Cordelia Schmid. Just ask: Learning to answer questions from millions of narrated videos. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 1686–1697, 2021. 2
- [43] Antoine Yang, Arsha Nagrai, Paul Hongsuck Seo, Antoine Miech, Jordi Pont-Tuset, Ivan Laptev, Josef Sivic, and Cordelia Schmid. Vid2seq: Large-scale pretraining of a visual language model for dense video captioning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10714–10726, 2023. 2
- [44] Eugene Yang, Andrew Yates, Kathryn Ricci, Orion Weller, Vivek Chari, Benjamin Van Durme, and Dawn Lawrie. Rank-k: Test-time reasoning for listwise reranking, 2025. 2
- [45] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. A survey on multimodal large language models. *National Science Review*, 11(12):nwae403, 2024. 2

- [46] Soyoung Yoon, Eunbi Choi, Jiyeon Kim, Hyeon-gu Yun, Yireun Kim, and Seung-won Hwang. List5: Listwise reranking with fusion-in-decoder improves zero-shot retrieval. *arXiv preprint arXiv:2402.15838*, 2024. 2
- [47] Emmanouil Zaranis, António Farinhas, Saul Santos, Beatriz Canaverde, Miguel Moura Ramos, Aditya K Surikuchi, André Viveiros, Baohao Liao, Elena Bueno-Benito, Nithin Sivakumaran, Pavlo Vasylenko, Shoubin Yu, Sonal Sannigrahi, Wafaa Mohammed, Ben Peters, Danae Sánchez Villegas, Elias Stengel-Eskin, Giuseppe Attanasio, Jaehong Yoon, Stella Frank, Alessandro Suglia, Chrysoula Zerva, Desmond Elliott, Mariella Dimiccoli, Mohit Bansal, Oswald Lanz, Raffaella Bernardi, Raquel Fernández, Sandro Pezzelle, Vlad Niculae, and André F. T. Martins. Movie facts and facts (mf²): A benchmark for long movie understanding, 2025. 2
- [48] Lu Zhang, Tiancheng Zhao, Heting Ying, Yibo Ma, and Kyusong Lee. Omagent: A multi-modal agent framework for complex video understanding with task divide-and-conquer. *arXiv preprint arXiv:2406.16620*, 2024. 2
- [49] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. *arXiv preprint arXiv:1904.09675*, 2019. 2, 4
- [50] Yaoyao Zhong, Wei Ji, Junbin Xiao, Yicong Li, Weihong Deng, and Tat-Seng Chua. Video question answering: Datasets, algorithms and challenges. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 6439–6455, 2022. 1
- [51] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. *arXiv preprint arXiv:2304.10592*, 2023. 2
- [52] Yutao Zhu, Huaying Yuan, Shuting Wang, Jiongnan Liu, Wenhan Liu, Chenlong Deng, Haonan Chen, Zheng Liu, Zhicheng Dou, and Ji-Rong Wen. Large language models for information retrieval: A survey. *arXiv preprint arXiv:2308.07107*, 2023. 2
- [53] Orr Zohar, Xiaohan Wang, Yann Dubois, Nikhil Mehta, Tong Xiao, Philippe Hansen-Estruch, Licheng Yu, Xiaofang Wang, Felix Juefei-Xu, Ning Zhang, et al. Apollo: An exploration of video understanding in large multimodal models. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 18891–18901, 2025. 1

A. Model Parameters

For Generator, we use Qwen2.5-VL 7B as the generator model, with the decoding parameters set to `temperature = 0.7` and `top_p = 0.9`.

For Reranker Training, we adopt LoRA Fine-tuning. The training dataset is from TREC dataset, containing a total 500 samples and we choose 90% of them for training and rest of them for evaluating. The reranker’s weights are set as $w_1 = 3$, $w_2 = 1.5$, and $w_3 = 1.2$. The training hyperparameters are as follows: `learning rate = 2e-6`, `batch size = 1`, and `epochs = 5`.

For the LoRA configuration, we set `lora_r = 16`, `lora_alpha = 32`, and `lora_dropout = 0.05`.

B. Prompts

In this section we provide the prompt variations used in [subsection 4.3](#) for Prompts 1-7 in [Figure 2](#), [Figure 3](#), [Figure 4](#), [Figure 5](#), [Figure 6](#), [Figure 7](#), and [Figure 8](#), respectively.

Answer the following question concisely in one sentence:
question: question

Figure 2. Prompt1

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.

question: question

Figure 3. Prompt2

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.
3. For the answer in number, you should answer the number (one, two, three, etc.) but not 1,2,3, etc. in the question.

question: question

Figure 4. Prompt3

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.
3. For the answer in number, you should answer the number (one, two, three, etc.) but not 1,2,3, etc. in the question.
4. You need to think about the question and answer it carefully.

question: question

Figure 5. Prompt4

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.
3. For the answer in number, you should answer the number (one, two, three, etc.) but not 1,2,3, etc. in the question.

the transcript of the video is: transcript
question: question

Figure 6. Prompt5

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.
3. For the answer in number, you should answer the number (one, two, three, etc.) but not 1,2,3, etc. in the question.
4. You need to think about the question and answer it carefully.
5. If the question is about the audio, you should answer the question based on the transcript of the video.
6. The transcript can be in different languages, you should answer the question in English.

the transcript of the video is: transcript
question: question

Figure 7. Prompt6

Answer the following question concisely in one sentence, you should follow the points:

1. You should answer the question as simple as possible, some questions may just need a word or two.
2. You don't need to answer the question in a very detailed way, just give a concise answer.
3. For the answer in number, you should answer the number (one, two, three, etc.) but not 1,2,3, etc. in the question.
4. The ASR transcript can be in different languages, you should answer the question in English.
5. Base your answer on both the ASR transcript and the video frames; if needed, use reasoning or general knowledge to give the most reasonable answer.

the ASR transcript of the video is: transcript
the question is: question

Figure 8. Prompt7